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Task 12

OSI MODEL

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Objective

To study the seven layers of the OSI model and map common protocols to each layer to understand network communication architecture.

OSI Layer Chart

Sr no.	Layer	Function	Examples
7	Application	Network services	HTTP, HTTPS
6	Presentation	Data formatting	JPEG, ASCII
5	Session	Manages sessions	RPC
4	Transport	Reliable delivery	TCP, UDP
3	Network	Routing and IP addressing	IP
2	Data Link	Frames and MAC addresses	Ethernet
1	Physical	Transmits bits	Cables, signals

Important Protocol Mapping

- HTTP/HTTPS: Layer 7 — Application
- TCP/UDP: Layer 4 — Transport
- IP: Layer 3 — Network
- Ethernet: Mainly Layer 2 — Data Link

Conclusion

The OSI model helps us understand network communication, troubleshoot network problems, and identify security risks at different layers. Understanding these seven layers builds a strong foundation for cybersecurity.

Interview questions

Q1. What is the OSI model?

The OSI model is a conceptual framework consisting of seven layers that explains how devices communicate over a network.

Q2. Which layer uses IP?

The Network layer, Layer 3.

Q3. Which layer uses TCP?

The Transport layer, Layer 4.